

ECE 210 Midterm 3 Worksheet

Note: This worksheet is not guaranteed to be entirely representative of the midterm's contents. Material may appear on this worksheet which will not appear on the midterm, and vice versa.

Partial Fraction Expansion

1) (Technique 1)

a) sub problem 1

Solution sub problem 1 solution

$$sF(s) = s \frac{s+3}{s(s+2)(s+5)} = \dots$$

b) sub problem 2

Solution sub problem 2 solution

2) (Technique 2)

a) sub problem 1

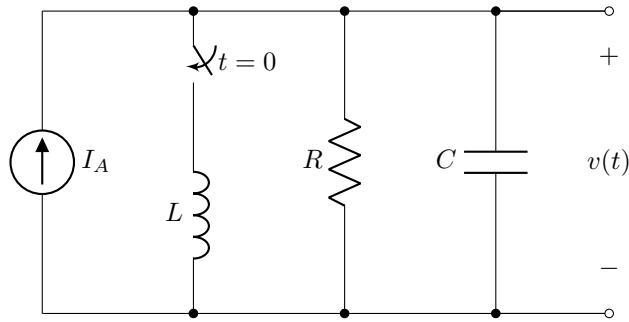
Solution sub problem 1 solution

b) sub problem 2

Solution sub problem 2 solution

S-Domain Circuit Analysis

3) In the circuit below, the switch has been open for a long time and is closed at $t = 0$. Let $I_A = 1$ mA, $L = 2$ H, $R = 1.5$ k Ω , and $C = \frac{1}{6}$ μ F.



- a) Find the initial conditions $i_L(0^-)$ and $v(0^-)$, and draw the s -domain equivalent circuit for $t \geq 0$.

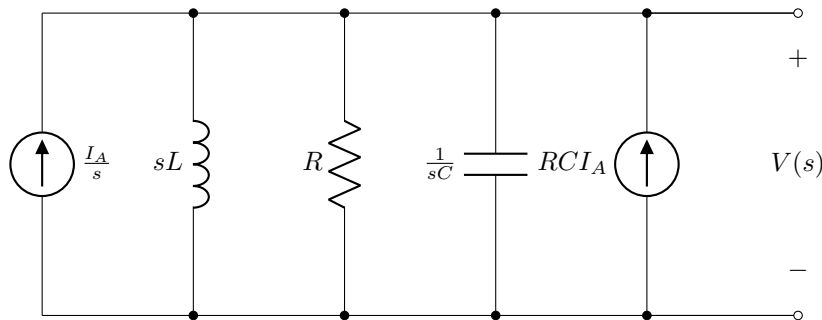
Solution With the switch open for a long time, the circuit is in DC steady state. The inductor branch is disconnected, so $i_L(0^-) = 0$. The capacitor is fully charged, and acts as an open circuit, so all of I_A flows through R :

$$v(0^-) = RI_A = (1.5 \text{ k}\Omega)(1 \text{ mA}) = 1.5 \text{ V}.$$

For $t \geq 0$, the switch is closed, and we redraw the circuit in the s -domain using element transformations. Each energy-storage element becomes its impedance plus a source that accounts for its initial condition. Since all the elements share the same two nodes, we use the parallel (Norton) form for each:

- **Current source:** $i_A(t) = I_A$ for $t \geq 0$, so it becomes $\frac{I_A}{s}$.
- **Resistor:** it is unchanged and stays R .
- **Inductor:** from $v_L(t) = L \frac{di_L}{dt}$ we get $V_L(s) = sL I_L(s) - L i_L(0)$, which gives $I_L(s) = \frac{V_L(s)}{sL} + \frac{i_L(0)}{s}$. This is an impedance sL in parallel with a current source $\frac{i_L(0)}{s}$. Because $i_L(0^-) = 0$, this source is zero, and the inductor is just sL .
- **Capacitor:** from $i_C(t) = C \frac{dv_C}{dt}$ we get $I_C(s) = sC V_C(s) - C v_C(0)$. This is an impedance $\frac{1}{sC}$ in parallel with a current source $C v_C(0)$. Here $C v(0^-) = C (RI_A) = RCI_A$.

Putting these together, the s -domain circuit has I_A/s , sL , R , $1/sC$, and the capacitor's RCI_A source all in parallel, with $V(s)$ across the output terminals. This is the circuit drawn below.



s-domain equivalent circuit for $t \geq 0$

- b) Find the zero-state and zero-input components of $V(s)$, and then find $v_{zs}(t)$ and $v_{zi}(t)$ for $t \geq 0$.

Solution The equivalent impedance seen by both sources is the parallel combination of sL , R , and $1/sC$:

$$Z_{eq}(s) = \left(\frac{1}{sL} + \frac{1}{R} + sC \right)^{-1} = \frac{sLR}{s^2 LRC + sL + R} = \frac{s/C}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}.$$

Zero-state (only the I_A/s source on):

$$V_{zs}(s) = Z_{eq}(s) \frac{I_A}{s} = \frac{I_A/C}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}.$$

Zero-input (only the RCI_A source on):

$$V_{zi}(s) = Z_{eq}(s) RCI_A = \frac{RI_A s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}.$$

Substitute values. Here $\frac{1}{RC} = 4000$, $\frac{1}{LC} = 3 \times 10^6$, $\frac{I_A}{C} = 6000$, and $RI_A = 1.5$. The denominator factors as

$$s^2 + 4000s + 3 \times 10^6 = (s + 1000)(s + 3000).$$

Zero-state, by partial fractions:

$$V_{zs}(s) = \frac{6000}{(s + 1000)(s + 3000)} = \frac{3}{s + 1000} + \frac{-3}{s + 3000}$$

$$\boxed{v_{zs}(t) = [3e^{-1000t} - 3e^{-3000t}] u(t) \text{ V}}$$

Zero-input, by partial fractions:

$$V_{zi}(s) = \frac{1.5 s}{(s + 1000)(s + 3000)} = \frac{-0.75}{s + 1000} + \frac{2.25}{s + 3000}$$

$$\boxed{v_{zi}(t) = [-0.75e^{-1000t} + 2.25e^{-3000t}] u(t) \text{ V}}$$

Marginal Stability & Resonance

4) problem 1

a) marginal stability part

Solution sub problem 1 solution

b) resonance part

Solution sub problem 2 solution

c) etc

Solution sub problem 3 solution

5) problem 2

a) marginal stability part

Solution sub problem 1 solution

b) resonance part

Solution sub problem 2 solution

c) etc

Solution sub problem 3 solution

Fourier Transforms

6) Find the Fourier transform:

$$g(t) = \frac{6}{t^2 + 9}$$

Solution This problem looks like the Fourier transform pair

$$f(t) = e^{-a|t|} \longleftrightarrow F(\omega) = \frac{2a}{a^2 + \omega^2}$$

with $a = 3$. However, we have $f(t)$, not $F(\omega)$ matching the right-hand side.

To take care of this, we can use the Symmetry Property, which states that if $f(t) \longleftrightarrow F(\omega)$, then

$$F(t) \longleftrightarrow 2\pi f(-\omega)$$

Now, with $a = 3$, we can evaluate: $f(-\omega) = e^{-3|-\omega|} = e^{-3|\omega|}$. Thus with the Symmetry Property,

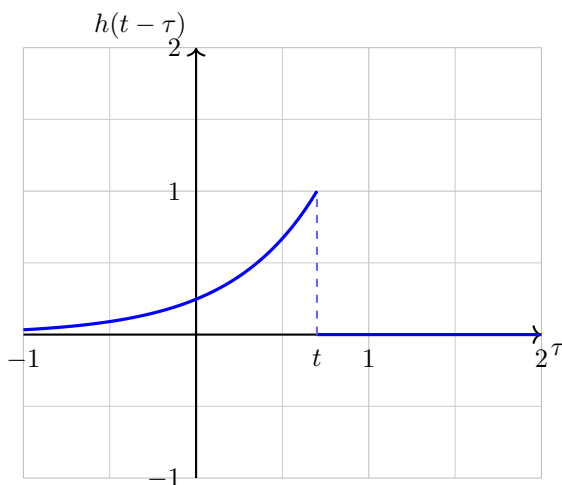
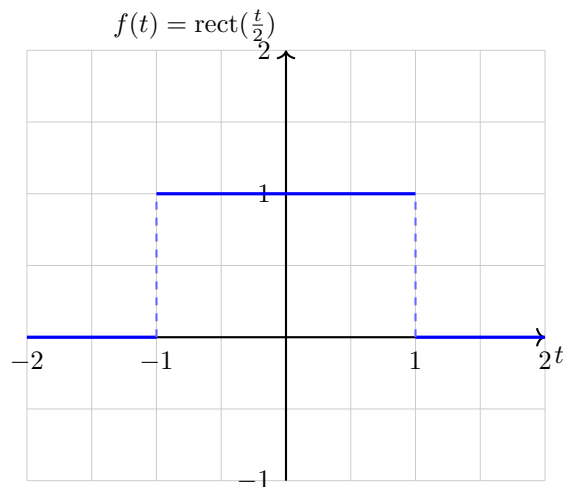
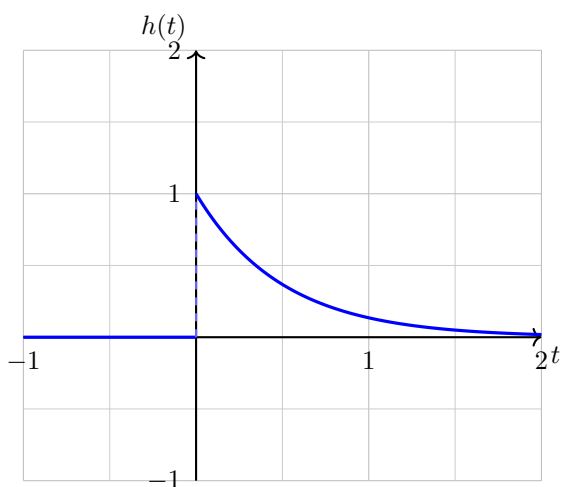
$$F(t) = \frac{2(3)}{3^2 + t^2} = \frac{6}{t^2 + 9} \longleftrightarrow 2\pi f(-\omega) = 2\pi e^{-3|\omega|}$$

Finally, since $g(t) = F(t)$, we can conclude that the Fourier transform is $G(\omega) = 2\pi e^{-|\omega|}$.

7 Find the output:

$$y(t) = e^{-2t}u(t) * \text{rect}\left(\frac{t}{2}\right)$$

Solution Let $h(t) = e^{-2t}u(t)$ and $f(t) = \text{rect}\left(\frac{t}{2}\right)$



$$y(t) = \int_{-\infty}^{\infty} h(t - \tau)f(\tau)d\tau$$

For $t < -1$: $y(t) = 0$ For $-1 \leq t < 1$:

$$\begin{aligned} y(t) &= \int_{-1}^t e^{-2(t-\tau)}d\tau = e^{-2t} \int_{-1}^t e^{2\tau}d\tau \\ &= e^{-2t} \cdot \left[\frac{1}{2}e^{2\tau} \right]_{-1}^t = e^{-2t} \left[\frac{1}{2}e^{2t} - \frac{1}{2}e^{-2} \right] = \frac{1}{2} \left[1 - e^{-2(t+1)} \right] \end{aligned}$$

For $t \geq 1$:

$$\begin{aligned} y(t) &= \int_{-1}^1 e^{-2(t-\tau)}d\tau = e^{-2t} \int_{-1}^1 e^{2\tau}d\tau \\ &= e^{-2t} \cdot \left[\frac{1}{2}e^{2\tau} \right]_{-1}^1 = e^{-2t} \left[\frac{1}{2}e^2 - \frac{1}{2}e^{-2} \right] = \frac{1}{2}(e^2 - e^{-2})e^{-2t} \end{aligned}$$

So,

$$y(t) = \begin{cases} 0, & \text{if } t < -1 \\ \frac{1}{2} [1 - e^{-2(t+1)}], & \text{if } -1 \leq t < 1 \\ \frac{1}{2}(e^2 - e^{-2})e^{-2t}, & \text{if } t \geq 1 \end{cases}$$